Analysis

The Real Numbers

The Absolute Value

Definition

We define the absolute value operator

$$|x| := \begin{cases} x \text{ if } x \geq 0 \\ -x \text{ if } x < 0 \end{cases}$$

Properties

- (Positive definite) $|a| \geq 0$. $|a| = 0 \iff a = 0$
- (Symmetry) $|a - b| = |b - a|$
- (Triangle Inequality) $|a + b| \leq |a| + |b|$

It then follows that:

- $x < y + \varepsilon \quad \forall \varepsilon > 0 \iff x \leq y$
- $x < y - \varepsilon \quad \forall \varepsilon > 0 \iff x \geq y$
- $x < \varepsilon \quad \forall \varepsilon > 0 \iff x = 0$

LUB and GLB

Let $S \subset \mathbb{R}$ be non-empty

- The set E is said to be bounded above if $\exists M \in \mathbb{R} \forall a \in E \quad a \leq M$
- A real number s is called the supremum of $E \iff$
  - s is an upper bound of E
  - $s \leq M$ for all upper bounds M of E.

If the number s, exists we write $s = \sup(E)$

Completeness Axiom

If $E \subset \mathbb{R}$ is nonempty and bounded above, E has a supremum.

Archimedian Principle

Given $a, b \in \mathbb{R}_+$ there is an integer $n \in \mathbb{N}$ such that $b < na$

Extended Real Numbers $\mathbb{R}^*$

$$\mathbb{R}^* = \mathbb{R} \cup \{-\infty, \infty\}$$

We then let

- $-\infty < a$,
- $a < \infty$,
- and $-\infty < \infty$ for all $a \in \mathbb{R}$.

Note that, for instance, the following are undefined: $\infty - \infty$, $0 \cdot \infty$, $\frac{\infty}{\infty}$, $\frac{a}{0}$

Supremums

We can redefine the supremum and infinum to work such that $\sup E = \infty \iff E$ is unbounded above and $\inf E = -\infty \iff E$ is unbounded below.

Injections, Surjections, and Bijections

Let function $f : X \to Y$, then

- f is one-to-one (or injective) if $x_1, x_2 \in X$ and $f(x_1) = f(x_2) \implies x_1 = x_2$
- f is onto (or surjective) if for every $y \in Y$ there exists some $x \in X$ such that $y = f(x)$

Functions that are both injective and surjective are also bijective.

Countability Definition

- E is said to be finite if either $E = \emptyset$ or if there exists a bijection $\{1, 2, ..., n\} \to E$, we say E is of size n.
- E is said to be countable if there exists a bijective function $f : \mathbb{N} \to E$
- E is at most countable if E is finite or countable.
- E is uncountable if E is neither finite or countable.

Countability Properties

- (Cantor) $(0, 1)$ is uncountable
- If S_1 and S_2 are countable, then $S_1 \cup S_2$ is countable (a similar theorem holds for being at most countable)
- If B is at most countable and $A \subset B$, then A is at most countable.
- If A is uncountable, and $A \subset B$, then B is uncountable.
- If A_1, A_2 are at most countable, then $A_1 \times A_2$ are at most countable.

Real Sequences

Limit of Real Sequences Definition

A real sequence (x_n) converges to a real number a if for every $\varepsilon > 0$, there is an $N \in \mathbb{N}$ such that

$$\forall n \geq N \text{ we have } |x_n - a| > \varepsilon.$$

The sequence is said to diverge to ∞ if for each $M \in \mathbb{R}$, there is $N \in \mathbb{N}$ such that

$$\forall n \geq N \text{ we have } x_n > M.$$

The sequence is said to diverge to $-\infty$ if for each $m \in \mathbb{R}$, there is $N \in \mathbb{N}$ such that

$$\forall n \geq N \text{ we have } x_n < m.$$

Limit Theorems

- Every convergent sequence is bounded
- Squeeze Theorem: If both $x_n \to a$ and $y_n \to a$ as $n \to \infty$, and if

$$x_n \leq a_n \leq y_n \text{ for all } n \geq N_0,$$

then, $a_n \to a$ as $n \to \infty$

- Let $E \subset \mathbb{R}$, if E has a finite supremum (i.e. E is non-empty and bounded above), then there is a sequence (x_n) with each $x_n \in E$ such that $x_n \to \sup E$ as $n \to \infty$. A similar statement holds if E has a finite infinum (i.e. E is non-empty and bounded below).
- If x_n and y_n converge (and if $\alpha \in \mathbb{R}$), then
  - $\lim_{n \to \infty} (x_n + y_n) = \lim_{n \to \infty} x_n + \lim_{n \to \infty} y_n$
  - $\lim_{n \to \infty} (x_n y_n) = \lim_{n \to \infty} x_n \cdot \lim_{n \to \infty} y_n$
  - $\lim_{n \to \infty} (\alpha x_n) = \alpha \lim_{n \to \infty} x_n$
  - If $\lim_{n \to \infty} y_n \neq 0$ and $y_n \neq 0$ for all $n \in \mathbb{N}$, then

$$\lim_{n \to \infty} \left(\frac{x_n}{y_n} \right) = \frac{\lim_{n \to \infty} x_n}{\lim_{n \to \infty} y_n}$$

- If $x_n \leq y_n$ for all $n \geq N$ for some $N \in \mathbb{N}$, then $\lim_{n \to \infty} x_n \leq \lim_{n \to \infty} y_n$

Monotone Sequences

A sequence is **monotone** if it is always increasing or decreasing.

A monotone sequence which is bounded must converge (we must only prove bounded above if it's increasing and vice-versa). An unbounded monotone sequence goes to either ∞ or $-\infty$ (an increasing sequence to ∞ and vice-versa.)

Subsequences

A subsequence of a sequence is a sequence where we only pick some terms from that sequence. Any subsequence of a convergent sequence is convergent, the same is not true for divergent sequences.

- There exists $t \in \mathbb{R}$ such that $\forall \varepsilon > 0$ there exists infinitely many $n \in N$ such that $|x_n - t| < \varepsilon \iff$ there exists a subsequence of (x_n) converging to t.
- The sequence (x_n) is not bounded above $\iff$ there exists a subsequence diverging to ∞.
- If a sequence converges to L, every subsequence also converges to L.

Infinite Series

Definitions

Let $S = \sum_{k=1}^{\infty} a_k$ be an infinite series with terms a_k. For each n the partial sum of order n is

$$s_n = \sum_{k=1}^{n} a_k$$

The series S converges to some $s \iff$

$$\lim_{n\to\infty} s_n = s.$$

Divergence Test

If $a_n \not\to 0$ as $n \to \infty$, then its series $S = \sum_{k=1}^{\infty} a_k$ diverges.

Boundedness

Suppose $a_k \geq 0$ for large k. Then, a_k's series converges $\iff$ the series of partial sums is bounded. That is, for some $M > 0$, $|s_n| \leq M$, for all $n \in N$.

Comparison Test

Suppose $0 < a_k \leq b_k$ for large k.

- $\displaystyle\sum_{k=1}^{\infty} b_k < \infty \implies \sum_{k=1}^{\infty} a_k < \infty$
- $\displaystyle\sum_{k=1}^{\infty} a_k = \infty \implies \sum_{k=1}^{\infty} b_k = \infty$

Integral Test

Suppose that $f : [1, \infty) \to \mathbb{R}$ is non-negative and decreasing on $[1, \infty)$. Let $a_k = f(k)$, $k = 1, 2, 3, \ldots$, then $\sum_{k=1}^{\infty} a_k = \sum_{k=1}^{\infty} f(k)$ converges $\iff \int_1^{\infty} f(x)\delta x < \infty$.

p-series Test

$$p > 1 \iff \sum_{k=1}^{\infty} \frac{1}{k^p} \text{ converges.}$$

Limit Comparison Test

Suppose that $0 \leq a_k, 0 \leq b_k$ for large k and that

$$L = \lim_{n\to\infty} \frac{a_k}{b_k} \text{ exists in } \mathbb{R}^*$$

- If $L \in (0, \infty)$, then

$$\sum_{k=1}^{\infty} a_k \text{ converges} \iff \sum_{k=1}^{\infty} b_k \text{ converges}$$

- If $L = 0$ and b_k's series converges, then a_k's series converges.
- If $L = \infty$ and b_k's series diverges, then a_k's series diverges.

Cauchy Condensation

Assume $(a_n)_{n\in\mathbb{N}}$ is a decreasing sequence with non-negative terms. The following are equivilent

1. The series $\sum_{k=1}^{\infty} a_n$ converges
2. The series $\sum_{k=1}^{\infty} 2^n a_{2^n}$ converges.

Absolute and Conditional convergence

- The series $S = \sum_{k=1}^{\infty} a_k$ converges absolutely if $\sum_{k=1}^{\infty} |a_k| < \infty$.
- The series S converges conditionally if S converges, but $\sum_{k=1}^{\infty} |a_k|$ diverges.

Absolute convergence implies conditional convergence, but not conversely.

Root Test, Limit Form

Let $a_k \in \mathbb{R}$ and assume $r = \lim_{k\to\infty} |a_k|^{\frac{1}{k}}$ exists. If,

- r < 1, then the series $\sum_{k=1}^{\infty} a_k$ converges absolutely
- r > 1, then the series diverges.

Ratio Test, Limit Form

Let $a_k \in \mathbb{R}$ and assume $r = \lim_{k\to\infty} \frac{|a_{k+1}|}{|a_k|}$ exists. If,

- r < 1, then the series $\sum_{k=1}^{\infty} a_k$ converges absolutely
- r > 1, then the series diverges.

Alternating-series Test

The series $\sum_{k=1}^{\infty} (-1)^k a_k$ is convergent $\iff$

- (a_k) is a decreasing sequence of non-negative numbers, and
- $(a_k) \to 0$ as $k \to \infty$.

Continuity

Definitions

Let f be a function and $f : \mathrm{dom}(f) \to \mathbb{R}$ where $\mathrm{dom}(f) \subseteq \mathbb{R}$ We say that f is continuous at $a \in \mathrm{dom}(f)$ if for any sequence (x_n) whose terms lie in $\mathrm{dom}(f)$ and which converges to a, we have $\lim_{n\to\infty} f(x_n) = f(a)$.

In short, f is continuous at $a \iff$ For any $\varepsilon > 0$, there exists some $\delta > 0$ such that

$$|x - a| < \delta \text{ and } |f(x) - f(a)| < \varepsilon$$

Properties

- Polynomials are continuous on $\mathbb{R}$.
- If $f, g : D \to \mathbb{R}$ are continuous on D, then so are
  1. αf
  2. $f + g$
  3. fg

Extreme and Intermediate Value Theorems

- (**Extreme Value Theorem**) Let $I \subset \mathbb{R}$ be a closed and bounded interval. Let $f : I \to \mathbb{R}$ be continuous on I. Then f is bounded on the interval I. Then, $m = \inf\{f(x) : x \in I\}$ has some $f(x_m) = m$ and same for sup
- Let $f : I \to \mathbb{R}$, where I is an open interval. If f is continous at a point $a \in I$, and $f(a) > 0$, then for some $\delta, \varepsilon > 0$, we have
$$f(x) > \varepsilon \forall x \in (a - \delta, a + \delta)$$
- (**Intermediate Value Theorem**) Let I be a non-degenerate interval and let $f : I \to \mathbb{R}$ be a continous function. If $a, b \in I, a < b$ then f attains on the interval (a, b), all values between $f(a)$ and $f(b)$. That is to say given y_0 between $f(a)$ and $f(b)$ there exists $x_0 \in (a, b)$ such that $f(x_0) = y_0$.
- (**Balzono's Theorem**) Let $f(x)$ be continuous on $[a, b]$ such that $f(a)f(b) < 0$, then there exists $c \in (a, b)$ such that $f(c) = 0$
- (**Rolle's Thoerem**) Let $f(x)$ be continous on $[a, b]$ and differentiable on (a, b) and $f(a) = f(b)$, then there exists some $c \in (a, b)$ such that $f'(c) = 0$
- If $f : [a, b] \to \mathbb{R}$ is a strictly increasing function, such that im f is an interval, then f is continuous on $[a, b]$

Limits of Functions

(**Right-hand limit**)Let $a \in \mathbb{R}$ and I be an open interval with left end-point a, then

$$\lim_{x\to a^+} f(x) = L \text{ if } \forall \varepsilon > 0 \exists \delta > 0 \text{ such that}$$

$$\text{if } a < x < a + \delta \text{ and } x \in I, \text{then } |f(x) - L| < \varepsilon$$

(**Limit comparisons**) $f(x) \leq g(x) \implies \lim_{x\to a} f(x) \leq \lim_{x\to a} g(x)$

Differentiability on $\mathbb{R}$

Definitions

A function $f : I \to \mathbb{R}$ is differentiable on $\mathbb{R}$ iff $f_{I'}(x) = \lim_{x\to a^I} \frac{f(x)-f(a)}{x-a}$ exists for all $a \in I$. A function f is continiuously differentiable on I if $f_{I'}$ exists and is continuous on I. Normal derivitive rules apply.

Mean Value Theorem

(**Mean Value Theorem**) If f is continuous on $[a, b]$ and differentiable on (a, b), then there is $c \in (a, b)$ with $a < b$, such that $f(b) - f(a) = f'(c)(b - a)$

(**More General MVT**) If f, g is continuous on $[a, b]$ and differentiable on (a, b), then there is $c \in (a, b)$ with $a < b$, such that $f'(c)(g(b) - g(a)) = g'(c)(f(b) - f(a))$.

Monotone Functions and Inverse Function Theorem

Definitions of monotonocity are similar for that of sequences.

(**Continuous bijections are monotone**) Let f be a $1 - 1$ function on I. Then f is strictly monotone on I and the inverse function is continuous and strictly monotone on $f(I)$.

(**Inverse function theorem**) Let f be a $1 - 1$ function, continuous on an open interval I. If $a \in f(I)$ and f' at the point $f^{-1}(a)$ exists and is non-zero, then f^{-1} is differentiable at a and

$$f^{(-1)'}(a) = \frac{1}{f'(f^{-1}(a))}$$

(**L'Hospital's Rule**) If $B = \lim_{x\to a, x\in I} \left[\frac{f'(x)}{g'(x)}\right]$ exists as an extended real number, then $\lim_{x\to a, x\in I} \left(\frac{f(x)}{g(x)}\right) = \lim_{x\to a, x\in I} \left(\frac{f'(x)}{g'(x)}\right)$

Taylor's Theorem

(**Taylor's Polynomial of f**) Let $n \in \mathbb{N}$ and $a < b$ be extended real numbers. If $f : (a, b) \to \mathbb{R}$ is a function differentiable n-times at a point $x_0 \in (a, b)$, then the taylor polynomial of degree n at x_0 is

$$P_n^{f,x_0}(x) = f_{x_0} + \sum_{k=1}^{n} \frac{f^k(x_0)}{k!}(x - x_0)^k$$

(**Lagrange Error Bound**)

$$R_n = \frac{M}{(n + 1)!}(x - x_0)^{n+1} \geq |f(x) - P_n^{f,x_0}(x)|$$

Algebra

Definitions

General Group Definition

- G1. (Closure) $*$ is an operation, so $g * h \in G$ for all $g, h \in G$.
- G2. (Associativity) $g * (h * k) = (g * h) * k$ for all $g, h, k \in G$.
- G3. (Identity) There exists an identity element $e \in G$ such that $e * g = g * e = g$ for all $g \in G$.
- G4. (Inverses) For every $g \in G$, there exists $g^{-1} \in G$ such that $g * g^{-1} = g^{-1} * g = e$.

If G is finite, we say the order of G is $|G|$, else we say the order of G is infinite.

Subgroups

A subset H of G is a subgroup $\iff$

- S1. (Non-empty) H is non-empty.
- S2. (Closure) $h, k \in H \Rightarrow hk \in H$
- S3. (Inverses) $h \in H \Rightarrow h^{-1} \in H$
- Alternatively, (S2) and (S3) $\iff h, k \in H \Rightarrow hk^{-1} \in H$.

Cyclic and Abelian Groups

- A generating element is typically denoted by g or $\langle g \rangle$ if it generates the group G. A generating element is an element such that if $h \in G$, then $h = g^n$ for some n. If an element of G generates all of G then G is cyclic. In fact, every element other than the identity in G is a generator.
- Abelian groups are groups with a commutative operation within the group. Note that some operations can be, in-general, non-commutative, but act commutative within a specific group, for instance, all groups are abelian in the trivial case ($G = \{e\}$)

Misc properties and elements

- The identity element is typically denoted as e or ε
- If $|G| = p$ for some prime p, then G is cyclic (Theorem 2.4.6).
- If a group is cyclic, then it is abelian.
- If $|G| < 6$, then it is abelian (Corollary 2.4.7).

Orders of elements

- $o(g)$ for $g \in G$ is called the "order of the element g" and is the smallest number which $g^n = e$. If $g^n \neq e$ for all $n \leq |G|$, then we say "the element g is of infinite order".
- In a finite group, all elements have finite order.
- In a finite group, $\forall g \in G. o(g) \mid |G|$
- In a finite group, if some $p \mid |G|$, then some $g \in G$ exists s.t. $o(g) = p$ (Cauchy's theorem)
- $o(g, h) = \text{lcm}(o(g), o(h))$

Common Groups and Their Properties

$\mathrm{GL}(n, \mathbb{R})$ and its subgroups

$\mathrm{GL}(n, \mathbb{R})$ is the General Linear Group of dimension n in $\mathbb{R}$. Generally, this is used to work with $\mathbb{R}^n$. Every invertable matrix of dimension n with real coefficients is a member of $\mathrm{GL}(n, \mathbb{R})$.

Useful Properties

- if $\det(A) \neq 0$, then some A^{-1} exists such that $\mathbb{A}^{-1} = I$.
- $\det(A)\det(B) = \det(AB)$
- $(AB)^{-1} = B^{-1}A^{-1}$
- $(AB)^T = B^T A^T$

$\mathbb{Z}/n\mathbb{Z}$ and $\mathbb{Z}/n\mathbb{Z}^\times$

- This is the additive group of order n under addition and multiplciation respectively, with the elements $\{0, 1, ..., n\}$
- For the multiplicitive case it is important that the order (n) is prime, otherwise it may fail to be closed.
- This group is isomorphic to all symmetry and permutative groups of the same order
- Euler's theorem:

$$a^{\varphi(n)} \equiv 1 \bmod n \text{ and to make FLT, } \varphi(p) = p - 1$$

Symmetries

D^n represents the Dihedral group (symmetries of a regular n-gon). In effect, this is e for the identity, h for a reflection, and g for a $\frac{360}{n}$ deg rotation, and each of their combinations. With $h^2 = e$ and $g^n = e$. For instance, the group for $n = 3$ is $\{e, g, g \cdot g, h, g \cdot h, h \cdot g\}$

Permutations

S_n represents the permutation group of size n. Consider $s \in S_n$, a mapping like $s : \{0, 1, ..., n-1\} \to \{0, 1, ..., n-1\}$. These mappings are often split into sets of loops, for instance, $(1234)(57)$ represents $1 \to 2 \to 3 \to 4 \to 1$ and $5 \to 7 \to 5$, with $6 \to 6$ being implied.

Cyclic groups and subgroups

Useful properties

Suppose that $H \leq G$ and H is finite, then

1. $|gH| = |H|$ for all $g \in G$
2. For a fixed $g \in G$, the number of $g_1 \in G$ s.t. $gH = g_1 H$ is equal to $|H|$

Lagranges Theorem

- If $H \leq G$, then $|H|$ divides $|G|$
- So if $g \in G$, then $o(g) \mid |G|$
- and for all $g \in G$, we have that $g^{|G|} = e$.

Group Morphisms

Assume G, H, and K are groups with operations $*$, $\cdot$, and $\otimes$ respectively.

$\ker \varphi :=$ Pre-image of ε for φ

Group Homomorphisms

A function $\varphi : G \to H$ is a group homomorphism iff

$$\varphi(x * y) = \varphi(x) \cdot \varphi(y)$$

Group Isomorphisms and Automorphisms

A group homomorphism φ that is also a bijection is an isomorphism. We write $G \cong H$.

Note that $\varphi^{-1} : H \to G$ then proves an isomorphism from H to G, and since we've already shown its existence, an isomorphism one way proves its existence both ways.

An isomorphism $G \to G$ is an automorphism of G.

Properties

- Assume H, K are subgroups of G with $H \cap K = \{e\}$, and $HK = \{hk \mid h \in H, k \in K\}$ If $hk = kh$ for all $h \in H$ and $k \in K$ (using the operation in G), then $H \times K$ is a subgroup in G, and $H \times K$ is isomorphic to HK.
- Assume H, K are **finite** subgroups of G with $H \cap K = \{e\}$, then $|HK| = |H| \times |K|$

Group Actions

Basic Definitions

Group Actions

- A (left) action of G on X is $G \times X \to X$, written $(g, x) \mapsto g \cdot x$ such that

$$g_1 \cdot (g_2 \cdot x) = (g_1 \cdot g_2) \cdot x \text{ and } e \cdot x = x$$

Take X = G,

- A right action is defined $g \cdot h = hg^{-1}$ for all $g \in G$ and $h \in X$.
- A conjugate action defined $g \cdot h = ghg^{-1}$ for all $g \in G$ and $h \in X$.

Orbits and Stabilizers

Let G act on X, and let $x \in X$. The stabilizer of x is defined to be

$$\mathrm{Stab}_G(x) = \{g \in G \mid g \cdot x = x\}$$

Let G act on X, and let $x \in X$. The orbit of x is defined to be

$$\mathrm{Orb}_G(x) = \{g \cdot x \mid g \in G\}$$

Properties

Orbit-Stabilizer theorem

Suppose G is a finite group acting on the set X, and let $x \in X$. Then,

$$|\mathrm{Orb}_G(x)| \times |\mathrm{Stab}_G(x)| = |G|$$

Misc

- Let G act on X,

$$x \sim y \Leftrightarrow y = g \cdot x \text{ for some } g \in G$$

defines an equivilence relation on G. The equivilence classes are the orbits of G.
- The Stabilizer of x is a subgroup of G.

Counting and Cayley

Polya Counting

(**Definition of Fixed set**) $\mathrm{Fix}(g) = \{x \in X \mid g \cdot x = x\}$

(**Polya Counting**) *the number of orbits in* $X = \frac{1}{|G|} \sum_{g \in G} |\mathrm{Fix}(g)|$

Conjugacy and the Class Equation

(**Conjugate Actions**) The action of a group on itself is called the conjugate action.

(**Conjugacy Class**) The orbits of the conjugacy action are the conjugacy classes.

(**Centralizer**) $C(g) := \{h \in G \mid gh = hg\}$

(**Centre of Group**) The center of the group G is $C(G) := \{g \in G \mid gh = hg \text{ for all } h \in G\}$. If $g \in C(G)$, g is "central".

(**Corollaries**)
- For all $g \in G$ the centralizer $C(g)$ is a subgroup of G
- $C(G)$ is a subgroup of G.
- If G is finite and $g \in G$, then $(\text{the number of conjugates of G}) \times |C(g)| = |G|$
- $\{e\}$ is always a conjugacy class of G.
- $\{g\}$ is a conjugacy class $\iff g \in C(G)$. Hence $C(G)$ is the union of all one-element conjugacy classes.

(**Class equation**) Say G is a finite group with conjugacy classes Conj_i, $i = 1, ..., k$.
1. $|\mathrm{Conj}_i|$ divides $|G|$.
2. $|G| = \sum_{i=1}^{k} |\mathrm{Conj}_i|$

(**Prime order fuckery**) If $|G| = p^k$, where p is prime, then $|G(G)| \geq p$ and every group G of order p^2 is abelian.

Cayley's Theorem

$$\mathrm{bij}(X) := \{\text{bijections } X \to X\}$$

- If G acts on the set X, then for all $g \in G$, the map $f_g : X \to X$ defined as $x \to g \cdot x$ is a bijection.

Let G be a group and let X be a set. Then
1. An action of G on X is equivilent to a group homomorphism $\varphi : G \to \mathrm{bij}(X)$
2. The action is faithful $\iff \varphi$ is injective.
3. If the action is faithful, then φ gives an isomorphism of G with $\operatorname{im} \varphi \leq \mathrm{bij}\ X$

(**Cayley's Thoerem**) Every finite group is isomorphic to a subgroup of a symmetric group.